

Advanced Econometrics – Lab 04

1 Ordered choice models

Exercise 1¹

We illustrate the ordered probit and logit techniques with a model of corporate bond ratings. The dataset `ratings.csv` contains information on 98 U.S. corporations' bond ratings and financial characteristics where the bond ratings are AAA (excellent) to C (poor). The integer codes underlying the ratings increase in the quality of the firm's rating, such that an increase in the response variable indicates that the firm's bonds are a more attractive investment opportunity. The bond rating variable (`rating83c`) is coded as integers 2-5, with 5 corresponding to the highest quality (AAA) bonds to the lowest. We model the 1983 bond rating as a function of the firm's income-to-asset ratio in 1983 (`ia83`: roughly, return on assets) and the change in that ratio from 1982 to 1983 (`dia`).

- a) Estimate ordered logit for `rating83c`.
- b) Are `ia83` and `dia` jointly significant?
- c) Interpret parameters' coefficients.
- d) Perform goodness-of-fit tests.
- e) Perform the Brant's test.
- f) Calculate and interpret pseudo- R^2 statistics.
- g) Calculate marginal effects and interpret them.

Exercise 2²

Let's analyse variable coding health status of respondents (`hstatus`). The variable takes three values: 1 for a person with bad health, 2 for moderate health, and 3 for good health status. Covariates for the dependent variable are: *income*, *female* – a binary variable (1 for women), *black*, and *num* – family size.

- a) Why shouldn't we use a logit/probit model for this problem?
- b) Estimate an ordered logit for health status with covariates mentioned above.
- c) Determine and interpret pseudo- R^2 statistics.
- d) Are variables in the model jointly significant?
- e) The parameter next to *income* is equal to $-3.39 * 10^{(-5)}$. Are we allowed to conclude on the basis of the value that *income* is insignificant?
- f) Having parameters estimated determine signs of marginal effects for *female* and *famsize* for the alternative *good health*.
- g) Perform goodness-of-fit tests.

¹On the basis of C.F. Baum, *An Introduction to Modern Econometrics Using Stata*, Stata Press, 2006, p. 257-258

²This exercise uses `randdata.dta` dataset from A. Colin Cameron and Pravin K. Trivedi, *Microeconometrics: Methods and Application*, Cambridge University Press, New York, May 2005.

- h) Determine and interpret marginal effects for the alternative *moderate health*.
- i) Determine and interpret marginal effects for the alternative *good health*.
- j) Verify hypothesis that *income* affects respondents' health linearly against the alternative that *income* has nonlinear, cubic impact. Use likelihood ratio test.

Exercise 3

A furniture retailer offers beds and mattresses. For every kind of bed a customer might choose out of three kinds of mattresses: ultra soft, moderate, and hard. The firm conducted a survey among its customers who decided to buy a bed with a mattress. An econometrician was asked to answer the question which customers choose what kind of mattresses. The econometrician proposed proportional odds model

$$mattress_i^* = \beta_1 age_i + \beta_2 age_i^2 + \beta_3 credit_card_i + \beta_4 female_i + \beta_5 instalments_i + \varepsilon_i, \quad (1)$$

where *mattress* equals to: 0 – ultra soft, 1 – moderate, 2 – hard. Variable *credit_card* was binary, and took value of 1 if a customer paid with their credit card. Binary variable *instalments* coded whether a customer bought a bed and a mattress by instalments.

Use $\alpha = 0.05$ significance level.

1. Why the econometrician decided to use proportional odds model?
2. Use `mattress.csv` to estimate ordered logit model.
3. Perform goodness-of-fit tests.
4. Calculate and interpret pseudo- R^2 statistics.
5. Test whether proportional odds ratio assumption is fulfilled.
6. Check joint significance of the variables in the model.
7. Which variables do have impact on the choice of customer?
8. Is there a relationship between age and softness of their mattress? Justify your answer.
9. How does customer's sex impact softness of mattress?
10. Determine marginal effects and interpret their values.
11. Apply general-to-specific approach for variable selection in the model.

Exercise 4

For data in `nel88.dta` file apply general-to-specific procedure to the model:

$$\begin{aligned} wage &= \beta_0 + \beta_1 age + \beta_2 race + \beta_3 married + \beta_4 grade + \beta_5 south + \beta_6 union + \\ &+ \beta_7 hours + \beta_8 ttl_exp + \beta_9 tenure + \beta' industry + \varepsilon \end{aligned}$$

Exercise 5

Import data from `Artificial_data.txt`. Apply general-to-specific approach for variable selection in the regression model for variable *y*. Present all the estimated models in one publication quality table.